

LONDON-CORBIN, KY, USA

WMO: 724243

Lat: 37.087N

Lon: 84.077W

Elev: 362

StdP: 97.05

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 03849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-8.7	-17.5	0.8	-8.7	-14.8	1.1	-6.6	9.7	8.3	8.7	7.1	1.9	290	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	32.5	22.6	31.3	22.5	30.3	22.1	24.6	29.6	24.0	28.8	23.4	28.0	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.6	26.6	22.5	18.0	26.1	22.1	17.5	25.7	76.0	29.9	73.5	28.9	71.4	28.0	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.1	5.8	-15.6	34.4	3.6	2.2	-18.2	36.0	-20.3	37.2	-22.3	38.5	-25.0	40.1		
			-16.1	25.6	3.5	0.9	-18.6	26.3	-20.7	26.9	-22.6	27.4	-25.2	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	2.2	4.2	8.5	14.0	18.5	22.6	24.3	23.8	20.6	14.1	8.1	4.1
	DBStd	9.11	6.53	5.76	5.77	4.91	3.98	2.77	2.14	2.36	3.56	4.66	4.99	5.67
	HDD10.0	856	253	177	98	19	2	0	0	0	0	15	94	198
	HDD18.3	2312	500	396	307	143	48	4	0	1	19	144	308	441
	CDD10.0	2244	11	14	52	139	266	378	443	429	317	143	36	15
	CDD18.3	658	0	0	2	13	54	132	184	172	86	14	1	0
	CDH23.3	5492	0	1	23	153	476	1082	1481	1389	740	141	7	0
	CDH26.7	1789	0	0	2	22	108	351	528	501	252	27	0	0
Wind	WSAvg	2.2	2.7	2.8	2.8	2.7	2.1	1.9	1.7	1.5	1.5	1.8	2.1	2.5
Precipitation	PrecAvg	1158	89	83	110	103	110	89	127	84	88	70	100	98
	PrecMax	1576	217	142	233	219	279	165	258	189	190	195	217	294
	PrecMin	855	27	16	32	15	28	18	28	20	22	0	10	10
	PrecStd	203	52	33	56	44	64	40	65	41	39	40	45	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	21.6	26.3	29.0	30.8	32.9	34.0	34.5	33.0	29.4	24.3	20.8
		MCWB	14.6	15.6	16.4	17.4	20.9	22.4	23.8	22.9	20.6	20.2	17.1	15.5
	2%	DB	17.3	18.9	23.2	27.0	29.1	31.5	32.2	32.4	31.3	26.9	21.2	17.8
		MCWB	13.4	13.4	14.8	16.6	20.0	22.5	23.5	22.8	21.1	18.8	14.8	14.1
	5%	DB	14.9	16.3	20.7	25.2	27.9	30.2	31.0	30.9	29.7	24.6	19.0	15.6
		MCWB	11.9	11.5	13.5	15.9	19.6	21.9	23.2	22.7	20.9	17.2	13.6	12.2
	10%	DB	12.5	13.8	18.2	23.1	26.3	28.7	29.7	29.6	27.8	22.5	16.9	13.5
		MCWB	9.4	9.7	12.3	15.1	19.1	21.3	22.7	22.3	20.1	16.7	12.4	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	16.5	17.4	19.5	22.9	24.8	25.7	25.2	23.9	22.2	18.1	16.6
		MCDB	18.2	20.2	23.7	25.5	27.7	30.0	30.7	30.9	28.8	26.6	21.8	19.4
	2%	WB	14.0	14.6	16.1	18.4	21.8	23.9	24.9	24.3	22.9	20.5	16.4	14.8
		MCDB	16.8	17.8	21.7	24.3	26.8	28.9	29.8	29.4	28.0	24.1	20.3	17.2
	5%	WB	12.5	12.4	14.8	17.3	20.9	23.1	24.1	23.7	22.0	19.0	14.8	13.0
		MCDB	14.8	15.3	19.4	22.4	25.9	27.9	28.7	28.5	26.6	22.7	17.7	15.0
	10%	WB	10.1	10.3	13.2	16.3	20.0	22.4	23.6	23.1	21.2	17.7	13.0	11.0
		MCDB	12.2	13.1	17.1	21.4	24.5	26.9	27.9	27.6	25.5	21.2	16.1	13.3
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.2	11.6	12.5	11.6	10.9	10.1	10.7	11.7	12.5	11.5	9.4
		MCDBR	11.9	13.6	14.7	15.4	13.6	12.5	11.8	12.3	14.3	15.0	14.3	12.0
		MCWBR	8.4	9.1	8.1	7.1	5.9	5.0	4.4	4.5	5.4	7.4	8.5	8.5
	5% WB	MCDBR	10.4	11.4	12.1	12.1	11.4	10.5	9.9	10.3	11.3	11.5	10.7	10.6
		MCWBR	8.0	8.6	7.6	6.5	5.6	4.9	4.2	4.2	4.7	6.4	7.6	8.4
Clear-Sky Solar Irradiance	taub	0.305	0.314	0.347	0.389	0.439	0.457	0.484	0.489	0.404	0.356	0.319	0.301	
	taud	2.508	2.476	2.422	2.301	2.183	2.168	2.115	2.087	2.311	2.441	2.513	2.533	
	Ebn at Noon	889	924	919	892	849	830	804	790	846	861	865	870	
	Edh at Noon	80	93	108	128	147	149	156	157	119	95	79	73	
All-Sky Solar Radiation	RadAvg	2.03	2.65	3.79	5.00	5.58	6.14	5.73	5.43	4.54	3.49	2.49	1.78	
	RadStd	0.21	0.37	0.34	0.36	0.49	0.34	0.31	0.29	0.47	0.45	0.28	0.18	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (22 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page